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Leveraging Transfer Learning for Accurate Detection of Diseases in Rice Leaf Images

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***Abstract*—A food source for billions of people worldwide, particularly in East and Southeast Asia, is rice. Unfortunately, several diseases, including bacterial blight, blast, brown spot, and tungro, can seriously reduce yields and often compromise the crop's ability to produce it. Using a dataset of 5,932 in-field images, this study explores the application of transfer learning to precisely identify and categorize these common rice leaf diseases. While pesticides may increase crop yields, they often degrade rice quality, which emphasizes the importance of early and accurate illness detection identification in order to assist farmers in minimizing losses. Using the most recent innovations in deep convolutional neural networks (DCNN), this study assesses the classification performance of five cutting-edge models: Xception, DenseNet121, InceptionResNetV2, ResNet50V2, and EfficientNetB0. The images underwent extensive pre-processing and data augmentation to enhance the quality and Variation of the training set, thereby improving model efficiency and generalization. With a 99.99% accuracy rate on the Mendeley dataset, the Xception model outperformed the other tested models. The results underscore the potential of transfer learning in agricultural disease diagnosis, offering a powerful tool for farmers to implement timely interventions and protect their crops from devastating losses.**

***Index Terms*—Deep Learning, Transfer Learning, Mendely Data, Xception, Rice Leaf Diseases.**

I. Introduction

About 50% of people in the planet eat rice (Oryza sativa) as a food source, particularly in East and Southeast Asia, where it serves as the primary source of calories [1], [2]. The global significance of rice as a food crop is underscored by its substantial contribution to the agricultural output of many countries, as well as its critical role in food security [3]–[5]. However, rice production faces significant challenges due to various diseases that can severely affect crop yield and quality [6]. Brown spot, blast, tungro, and bacterial blight are a few of the most common and harmful diseases that affect rice leaves. In addition to endangering rice crop output, these illnesses cause farmers to suffer significant financial losses, especially in areas where rice cultivation is a primary source of income.

Proper identification of these diseases is crucial for effective crop management. Early diagnosis allows farmers to implement appropriate treatments, thereby minimizing crop damage and financial loss. Traditionally, the detection and diagnosis of rice diseases have relied on manual inspection by experts, which is time-consuming, subjective, and often inaccessible to farmers in remote areas. This highlights the need for automated, accurate, and scalable disease detection methods.

In recent years, DL, particularly DCNNs [7], has emerged as a powerful tool for image classification tasks [8], including the diagnosis of plant diseases. DCNNs have demonstrated remarkable accuracy in classifying images across various domains, such as healthcare, automotive, and agriculture [9]–[17]. In the context of plant pathology, DCNNs have been successfully applied to the classification of diseases in crops like apples, tea, and tomatoes [18]. This paper explores the application of DCNNs to the problem of rice leaf disease detection, leveraging transfer learning techniques to enhance model performance.

Transfer learning allows the adaptation of pre-trained models to new tasks with limited data, significantly improving the efficiency and accuracy of model training [19]. By utilizing pre-trained models, we can leverage the knowledge acquired from large, diverse datasets, thereby reducing the need for extensive labeled data specific to rice diseases. This approach is particularly advantageous in agricultural settings where labeled data may be scarce or expensive to obtain.

This research, we evaluated five state-of-the-art DCNN architectures—Xception, ResNet-50 V2, InceptionResNetV2, DenseNet-121, and EfficientNetB0—in classifying leaf diseases of rice using a dataset of 5932 on-field images. These models have been fine-tuned and augmented with additional

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layers, such as global average pooling, AlphaDropout, and dense layers, to enhance their accuracy and generalization capabilities. The training data has been made more diverse and high-quality through the data preparation and augmentation method, which lowers the likelihood of overfitting [20]–[24].

Our experiments demonstrate that the Xception model, in particular, achieved an impressive 99.99% accuracy in classifying rice leaf diseases, outperforming the other architectures on the Mendely dataset. This exceptional performance underscores the potential of transfer learning in agricultural applications, providing a robust solution for the accurate and efficient diagnosis of rice diseases in various farming contexts.

This paper makes the following primary contribution.

1) Evaluated Xception, ResNet-50 V2, InceptionResNetV2, DenseNet-121, and EfficientNetB0 for rice disease classification.
2) Applied extensive data augmentation and preprocessing to improve model robustness and reduce overfitting.
3) Introduced additional layers to the models, improving accuracy and generalization.
4) Demonstrated Xception's superior performance in disease classification.
5) Conducted a detailed comparison of model performance across multiple architectures.
6) Efficiently adapted pre-trained models to enhance rice disease detection.

II. State of the Art in the Field

In the realm of rice crop diseases and pest detection [25], the primary focus lies on the utilization of deep learning models. In order to highlight early and accurate categorization, the author employed a variety of models, including ResNet50V2, ResNet101V2, VGG16, and VGG19. Using digital image processing, data augmentation, cross-validation background removal, and leaf segmentation approaches, this work achieves an accuracy of 86.79% for actual data and 75% for fictional data. Using machine learning [26] techniques such as KNN, Decision Tree, Naïve Bayes, and Logistic Regression, diseases affecting rice leaves such as brown spot, bacterial leaf blight, and leaf smut can be detected early. Using 10-fold cross validation, this work attains 97% accuracy. Rice crop diagnosis using weighted average ensemble learning algorithms [27]. Using data augmentation and preprocessing approaches, the author used a Kaggle dataset. The suggested DECNN model achieves a 98.33% accuracy using a variety of deep learning approaches, including InceptionV3, InceptionResNetV2, DenseNet121, DenseNet169, and DenseNet201.Image processing techniques, combined with methods such as Otsu's Segmentation, LBP, HOG, and SVM, have been utilized for detecting rice leaf diseases [28]. These techniques yielded an accuracy rate of 94.6% in the classification of three types of rice leaf diseases: brown spot, bacterial leaf blight, and leaf smut. In a different investigation on recognizing of rice leaf diseases [29], a GAN-based enhanced data pipeline is the subject. With the addition of an SG2-ADA and variation of the Laplacian filter in this work, the four primary illnesses affecting rice are more accurately detected by the SSD and Faster-RCNN.

III. Methodology

A. Dataset

In our study, we utilized a dataset provided by [30], which consists of 5932 images across four disease categories: Bacterial Blight with 1584 images, Blast with 1440 images, Brown Spot with 1600 images, and Tungro with 1308 images. This dataset is accessible through the Mendeley archive.

B. Data Pre-processing

Preprocessing is a crucial initial step in image processing [31]–[34]. The dataset were first gotten from the Mendeley database. In this stage, image preprocessing techniques were applied to standardize input dimensions, find the diversity of the training set, and increase the model's accuracy to generate new images. Techniques such as image scaling, normalization, and data augmentation were utilized. Preprocessing also increased the contrast of the leaf images by eliminating unnecessary noise. During the training phase, these methods help the model learn a wide range of patterns while minimizing the risk of overfitting [35].

C. Data Augmentation

Using Python's ImageDataGenerator package, augmentation techniques were applied to the rice deficiency datasets as described in [36]–[38]. Each image in the Mendeley dataset was resized to a resolution of 300 × 300 pixels. Various augmentation methods were employed for images in each class, including fill mode, horizontal and vertical flips, rotation without automatic cropping, rescaling, width and height shifts, shear ranges, and zoom effects. The specific parameters for these augmentations are presented in Table I. Following supplementation, the increased Mendeley dataset contained 5932 total photos.

TABLE I
Data Augmentation

Content	Value
Rescale	1./255
Range of Rotations	10
Range of width shift	0.2
Range of height shift	0.2
Shear length	0.2
Zoom extent	0.2
Flipping Horizontally	True
Vertical reversal	False
Fill mode	Nearest

D. Proposed Framework

The proposed methodology, depicted in Fig-1, involves four key stages for classifying rice leaf diseases. The process begins with acquiring the rice leaf image dataset, followed by data preprocessing to improve and standardize image quality. Subsequently, the dataset is enriched through data augmentation techniques, which increase its diversity and robustness. In this

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study, various transfer learning models such as ResNet50v2, DenseNet121, Xception, InceptionResNetV2, and EfficientNetB0 are employed for feature extraction. After freezing some pre-trained layers, additional layers such as GlobalAveragePooling2D, AlphaDropout, BatchNormalization, Dense, Flatten, and a softmax classifier are added for the classification of four rice leaf diseases. These additional layers enhance feature extraction, reduce overfitting, and support efficient classification. The combination of preprocessing, data augmentation, transfer learning, layer addition, and the softmax classifier ensures a reliable and accurate model for classifying rice leaf diseases.

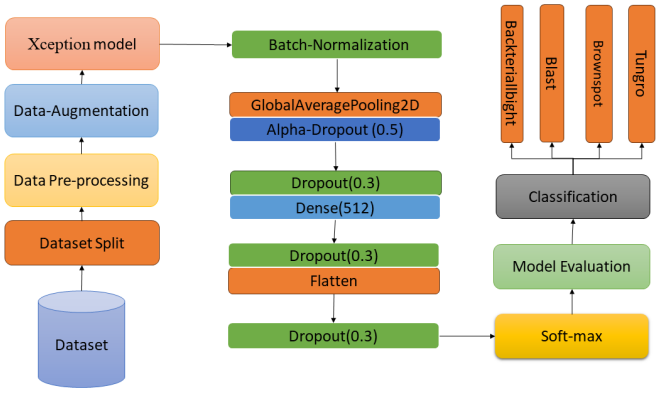


Fig. 1. System Architecture

E. Proposed Model

Using the pre-trained Xception model on the ImageNet dataset [39], rice leaf images were categorized in this study. The first layer up to the Global Average Pooling2D layer was used to extract features, also known as bottleneck features. Using AlphaDropout as the second layer helped to lessen overfitting and increase model generalization. This layer introduces noise during training while maintaining the expected activation value. A basic overview of the proposed models is illustrated in Fig. 2. The training parameters used in the experiment are detailed in Table II.

TABLE II
TRAINABLE PARAMETER LIST

Model	Total parameter	Trainable parameter	Non-Trainable parameter
Xception	21912620	21858092	54528
DenseNet121	7564356	7480708	8364
InceptionResNetV2	55125732	55065181	60544
ResNet50V2	24615940	24570500	45440
EfficientNetB0	4707495	4665472	42023

In this work, the prepared model's softmax layer was replaced with a new randomized-weight soft max layer using transfer techniques. Next, in order to identify the malware photos, the new softmax layer was trained using the bottleneck characteristics [40], [41].

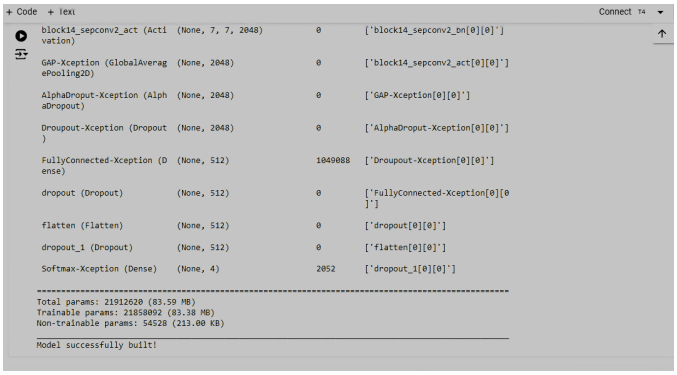


Fig. 2. Proposed model description

F. Transfer Learning and Hyperparameter Setting

Within the discipline of machine learning, transfer learning makes use of connections between functions, models, or outcomes from other domains in order to move knowledge from one source model to another [42]. Here, an extensive dataset-pre-trained model, such as ImageNet, is utilized to achieve accurate classification on a smaller dataset. This approach addresses the challenge of deep learning models struggling to achieve high accuracy when starting with a small dataset, as the model may not capture sufficient details about the data's variability [43]. Bottleneck features are extracted from the initial layers up to the Global Average Pooling2D layer. To mitigate overfitting, AlphaDropout is employed during training, introducing noise, while the softmax layer handles image classification [44]. The training hyper-parameter values are detailed in Table III.

TABLE III
TRAINING HYPERPARAMETER VALUE

Name	Value
Batch size	32
Alpha-Dropout	0.5
Dropout	0.3
Dense	512
Epoch	50
Learning rate	0.0001
Optimizer	Adam
Activation	Soft-max
Loss function	Categorical cross-entropy

Key parameters significantly influence the performance of CNNs and other deep learning models. Hyperparameters play a crucial role in generating meaningful results before the training process begins. For our proposed model, we utilized the hyperparameters listed in Table III.

IV. OUTCOMES AND INTERPRETATION

In our study, we developed a transfer learning based on the dividing of rice leaf diseases using the Mendely dataset, which consists of 5932 images and four classes.

The classification result [45] obtained by using the test datasets of rice leaf diseases is shown in Fig. 3 Utilizing 597 images from the four classes Bacterialblight, Blast, Brownspot,

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and Tungro, the categorization accuracy was ascertained. Correctly diagnosed cases are shown in the confusion matrix figures as 100% TPR and highlighted blocks along the diagonal path. Blocks outside of this bound are seen as signs of incorrect categorization or misdiagnosis.

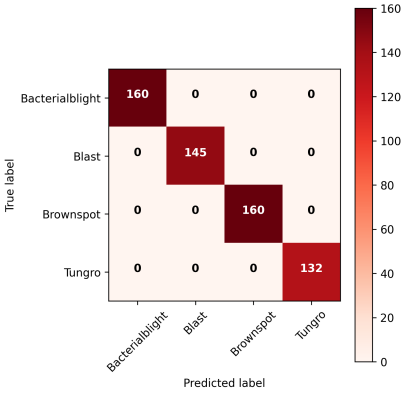


Fig. 3. Confusion matrix using test data

In our work, we use five different modified deep transfer learning learning model and the best model Xception model. Table IV shows the five deep learning model accuracy, precision, recall, and f-1 score.

TABLE IV
PERFORMANCE COMPARISON OTHERS STATE-OF-THE-ART MODELS

Model	Accuracy (%)	Precision (%)	Recall (%)	F-1 Score (%)
Proposed Model	99.99	99.99	99.99	99.99
DenseNet121	98.99	90.01	99.06	99.02
InceptionResNetV2	92.46	93.91	91.69	92.04
ResNet50V2	86.59	90.16	85.57	85.96
EfficientNetB0	74.06	74.38	75.67	74.72

The accuracy of the model was 99.99% but the others was DenseNet121 was 98.99%, InceptionResNetV2 was 92.46%, ResNet50V2 was 86.59%, and EfficientNetB0 was 74.06% respectively using the test data. In Fig. 4, 5 and 6 all of the updated transfer learning model's training and validation accuracy,loss, and convergence curves overlap after 50 epochs.

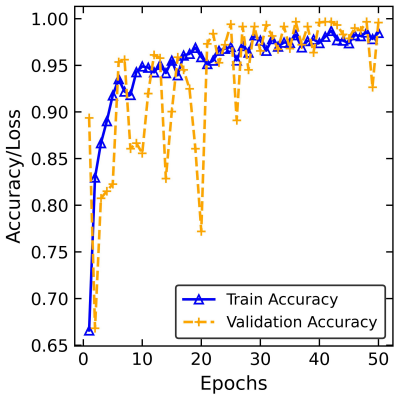


Fig. 4. Accuracy curve on test data

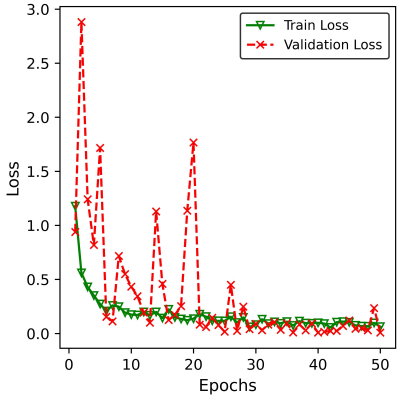


Fig. 5. Training Loss curve on test data

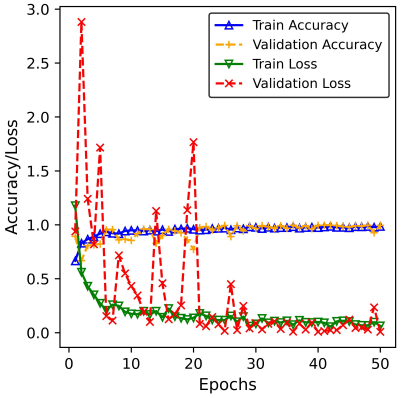


Fig. 6. Training Convergence curve on test data

The precision-recall curve is shown in Fig. 7 for assessing the rates of false positives (FP) and false negatives (FN) at various levels, particularly when test samples from different classes are unbalanced [46]. The Xception model's overall AUC in the test data was 100%, with the greatest AUC seen being 100% across all classes. It is clear that the AUC values for all classes are 1, as shown in Fig. 8, the ROC curve [47] of the Xception model comparing the performance of the four classes' rice leaf diseases with the test dataset. Table V Our model performance in comparision to the state-of the -art -model.

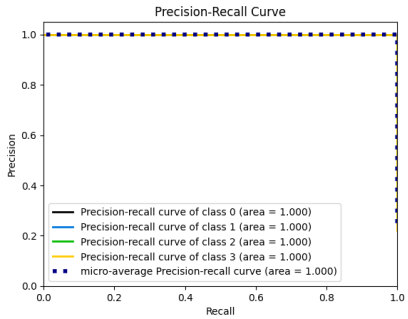


Fig. 7. Precision-Recall curve using test data

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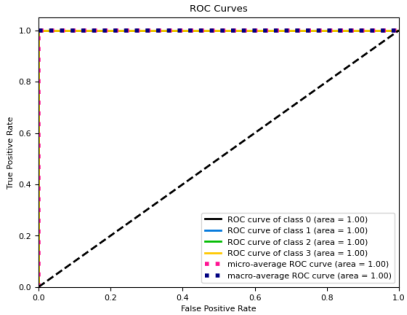


Fig. 8. ROC Curve on test data

TABLE V
COMPARISON WITH PREVIOUS STUDY

Model	Accuracy
CNN-SVM Hybrid [48]	97%
ResNet50V2 [49]	98.38%
Deep Learning [50]	99.66%
Our Study	99.99%

V. CONCLUSION AND FUTURE WORK

Four rice diseases—Bacterial Blight, Blast, Brown Spot, and Tungro—are classified using five pre-trained models in this work. Global average pooling2D, AlphaDropout, dropout, flatten, and dense layers have all been added to the models. DenseNet121, InceptionResNetV2, ResNet50V2, EfficientNetB0, and Xception are the names of the improved models. The application of several preprocessing and augmentation techniques also increased the amount of data points and decreased overfitting. The algorithms demonstrated varying levels of accuracy in predicting rice leaf diseases, with the proposed model achieving the highest performance, reaching 99.99% accuracy on the test data. This highly accurate model could greatly benefit farmers by enabling early disease detection, potentially improving rice yields. Additionally, it offers cost and time savings by reducing the reliance on specialists. We intend to use this dataset to investigate the efficacy of ensemble learning strategies in subsequent research.

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